A MAXIMAL IDEAL OUTSIDE AN OKA FAMILY OF IDEALS IS PRIME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Oka condition on a family of ideals. Applied at the last line of the counted proof (line 5432). Turns the two strictly larger ideals back into members of the family, which yields the contradiction.
- The ideal quotient and the ideal generated with an element. Used in the counted proof at lines 5429-5430. Produces two ideals strictly containing the given one from a pair of non-members with product inside it.

Proposition 0.1. *If $\mathcal{F}$ is an Oka family of ideals, then any maximal element of the complement of $\mathcal{F}$ is prime.*

Proof. Suppose $I \notin \mathcal{F}$ is maximal with respect to not being in $\mathcal{F}$ but I is not prime. Note that $I \neq R$ because $R \in \mathcal{F}$. Since I is not prime we can find $a, b \in R - I$ with $ab \in I$. It follows that $(I, a) \neq I$ and $(I : a)$ contains $b \notin I$ so also $(I : a) \neq I$. Thus $(I : a), (I, a)$ both strictly contain I, so they must belong to $\mathcal{F}$. By the Oka condition, we have $I \in \mathcal{F}$, a contradiction. $\square$

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